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10 August 1965
SUPERSEDING
MIL-R-8202C(USAF)
5 June 1957

MILITARY SPECIFICATION

REGULATOR, OXYGEN, DILUTER DEMAND D-2A
(REGULATOR, AUTOMATIC DILUTER DEMAND, OXYGEN, PRESSURE BREATHING, TYPE D-2A)

1. SCOPE

1.1 This specification covers one type of low pressure (50 to 500 pounds per square inch) pressure breathing, oxygen, automatic diluter demand regulator designated D-2A.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

Federal

BB-N-411	Nitrogen, Technical
PPP-B-566	Boxes, Folding, Paperboard
PPP-B-585	Boxes, Wood, Wire-bound
PPP-B-591	Boxes, Fiberboard, Wood-Cleated
PPP-B-601	Boxes, Wood, Cleated-Plywood
PPP-B-621	Boxes, Wood, Nail and Lock-Corner
PPP-B-636	Box, Fiberboard
PPP-B-665	Boxes; Paperboard, Metal Stayed (Including Stay Material)
PPP-B-676	Boxes, Set-Up, Paperboard
PPP-T-60	Tape; Pressure-Sensitive Adhesive, Waterproof, for Packaging

Military

MIL-P-116	Preservation, Methods of
MIL-C-5541	Chemical Films for Aluminum and Aluminum Alloys
MIL-P-7105	Pipe Threads, Taper, Aeronautical National Form, Symbol ANPT, General Specification for
MIL-S-7742	Screw Threads, Standard, Optimum Selected Series: General Specification for
MIL-P-7788	Plate, Plastic, Cockpit and Interior Controls Lighting
MIL-A-8625	Anodic Coatings, for Aluminum and Aluminum Alloys
MIL-L-10547	Liners, Case, Waterproof
MIL-L-25567	Leak Test Compound, Oxygen Systems
MIL-G-25728	Gage, Pressure, Dial Indicating, Oxygen, 0-500 PSI, General Specification for
MIL-M-26512	Maintainability Requirement for Aerospace Systems and Equipment
MIL-R-26667	Reliability and Longevity Requirements, Electronic Equipment, General Specification for
MIL-O-27210	Oxygen, Aviator's Breathing, Liquid and Gas

FSC 1660

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MIL-T-38180	Tester, Leakage Flow Demand, Oxygen Regulator TTU-213/E
MIL-D-70327	Drawings, Engineering and Associated Lists

STANDARDS

Military

MIL-STD-129	Marking for Shipment and Storage
MIL-STD-130	Identification Marking of U.S. Military Property
MIL-STD-143	Specifications and Standards, Order of Precedence for the Selection of
MIL-STD-1186	Cushioning, Anchoring, Bracing, Blocking, and Waterproofing; with Appropriate Test Methods
MS20913	Plug, Square Head Pipe Thread
MS25453	Lights, Panel-Information Plate, Plate Supporting Type
MS33558	Numerals and Letters, Aircraft Instrument Dial, Standard Form of
MS33586	Metals, Definition of Dissimilar

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Preproduction. This specification makes provisions for preproduction testing.

3.2 Selection of specifications and standards. Specifications and standards for necessary commodities and services not specified herein shall be selected in accordance with MIL-STD-143.

3.3 Materials

3.3.1 Nonferrous materials. Except where ferrous materials are essential, nonferrous materials shall be used for all parts of the regulator.

3.3.2 Nonmetallic materials. Any nonmetallic material that is easily deteriorated or otherwise easily affected adversely by continued service use with oxygen shall not be used.

3.3.3 Elastomers. All elastomers of natural rubber or any synthetic rubber other than certain silicone types (which are to be approved by the procuring activity) shall have been manufactured not more than 12 months prior to the date the regulator was manufactured. The cure date (quarter and year) shall be stamped on the elastomer component, provided there is sufficient area. Cure dates and age limitations shall not apply to the silicone types. The demand diaphragm shall be fabricated of an approved silicone compound or silicone-coated cloth. All elastomers shall be free from foreign agents that might cause objectionable or intolerable odors.

3.3.4 Protective treatment. When materials that are subject to corrosion in oxygen, salt air, or any other atmospheric conditions likely to occur during service usage are used in the construction of the regulator, they shall be protected against such corrosion in a manner that will in no way prevent compliance with the performance requirements of this specification. The use of any protective coating that will crack, chip, or scale with age or extremes of atmospheric conditions shall be avoided.

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3.3.5 Plugs. All openings in the regulator shall be closed with caps or plugs to prevent dust and any other foreign matter from entering the regulator during shipment and storage. If threaded plugs are used, they shall conform to MS20913, except that the material shall be a plastic approved by the procuring activity. The material and the design of any other caps or plugs shall be subject to the approval of the procuring activity.

3.3.6 Dissimilar metals. Unless suitably protected against electrolytic corrosion, dissimilar metals shall not be used in intimate contact with each other. Dissimilar metals are defined in MS33586.

3.4 Design. The regulator shall have a lighted panel, shall be of the automatic pressure breathing diluter demand type, and shall be designed to operate on oxygen in the pressure range of 50 to 500 pounds per square inch. The dimensions of the regulator shall be as specified on figures 1, 2, 3, and 4. A suitable mask pressure relief valve shall be incorporated in the body of the regulator.

3.5 Construction. The regulator shall be constructed so that no parts will work loose in service. It shall be built to withstand strains, jars, vibrations, and any other conditions incident to shipping, storage, installation, and service.

3.5.1 Air port. The regulator shall have an air port for mixing the varying quantities of air with oxygen.

3.5.1.1 Diluter lever. The regulator shall have a diluter lever for manually shutting off the air port and for allowing the regulator to deliver pure oxygen. Stops shall limit the motion of the diluter lever at the "normal oxygen" and "100 percent oxygen" positions so that when the lever is pushed to the "normal position" stop, the normal dilution characteristics will be maintained. When the diluter is pushed to the "100 percent oxygen" stop, the regulator shall deliver pure oxygen at all altitudes. Tension of the diluter shall be great enough so that there will be no possibility of a change in setting because of vibration of the aircraft.

3.5.1.2 Air port screen. A screen in the air port shall prevent foreign particles from entering the regulator. The screen shall be not finer than 100 mesh nor coarser than 30 mesh.

3.5.1.3 Diaphragm. All diaphragms used in the regulator shall be fabricated of a silicone compound, silicone-coated cloth, or any other material specifically approved by the procuring activity.

3.5.2 Safety pressure. An automatic safety pressure shall prevent negative pressure within the oxygen mask at altitudes between 30,000 and 40,000 feet.

3.5.3 Emergency pressure control. An emergency pressure control shall apply mechanical pressure on the diaphragm of the regulator so that a positive pressure can be obtained at the outlet of the regulator at altitudes where automatic positive pressure is not delivered by the regulator. This control shall be located on the regulator panel as specified on figure 1. The control toggle shall have three positions. When the control toggle is pushed

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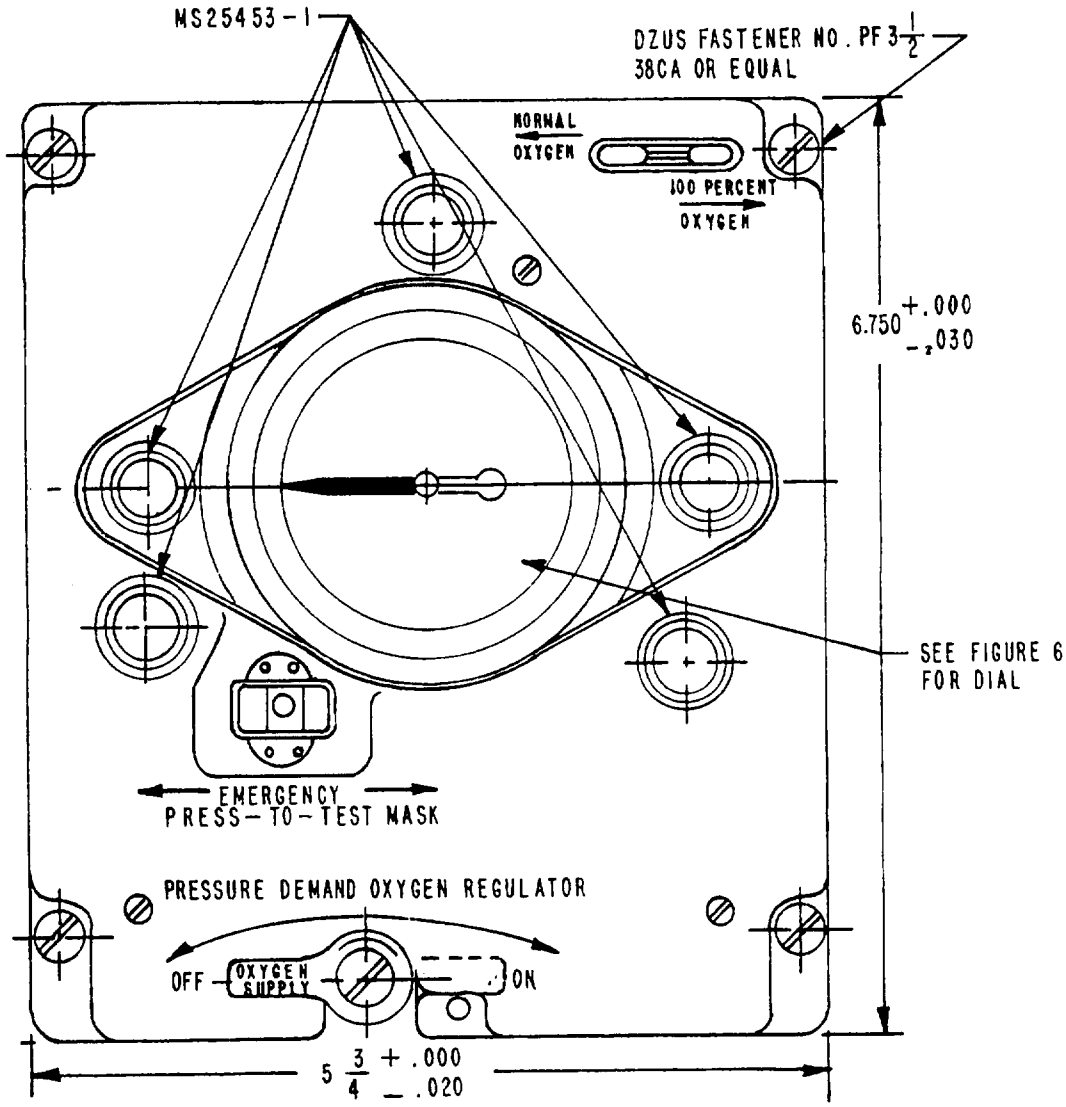


FIGURE 1. Regulator, Front View

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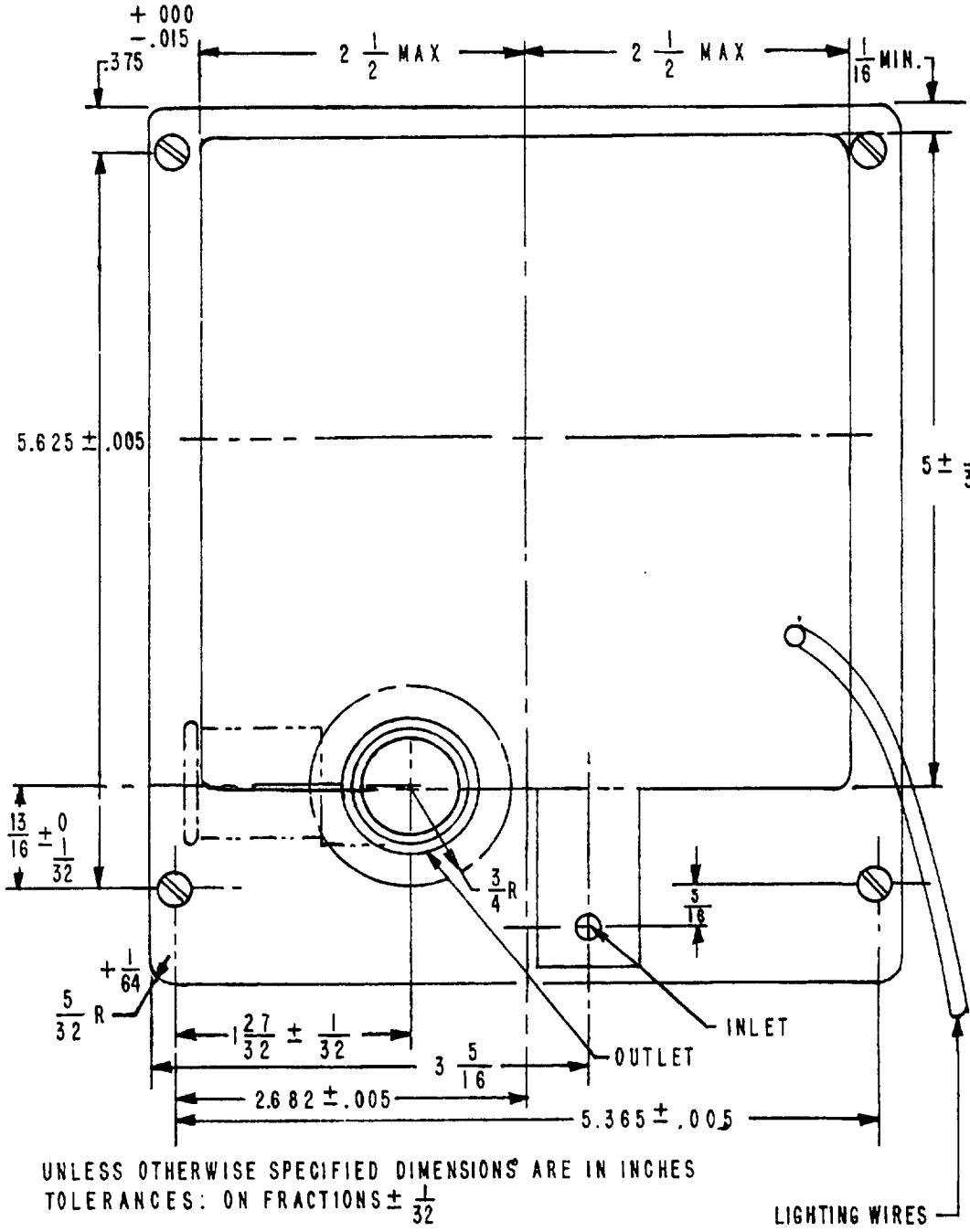


FIGURE 2. Regulator, Rear View

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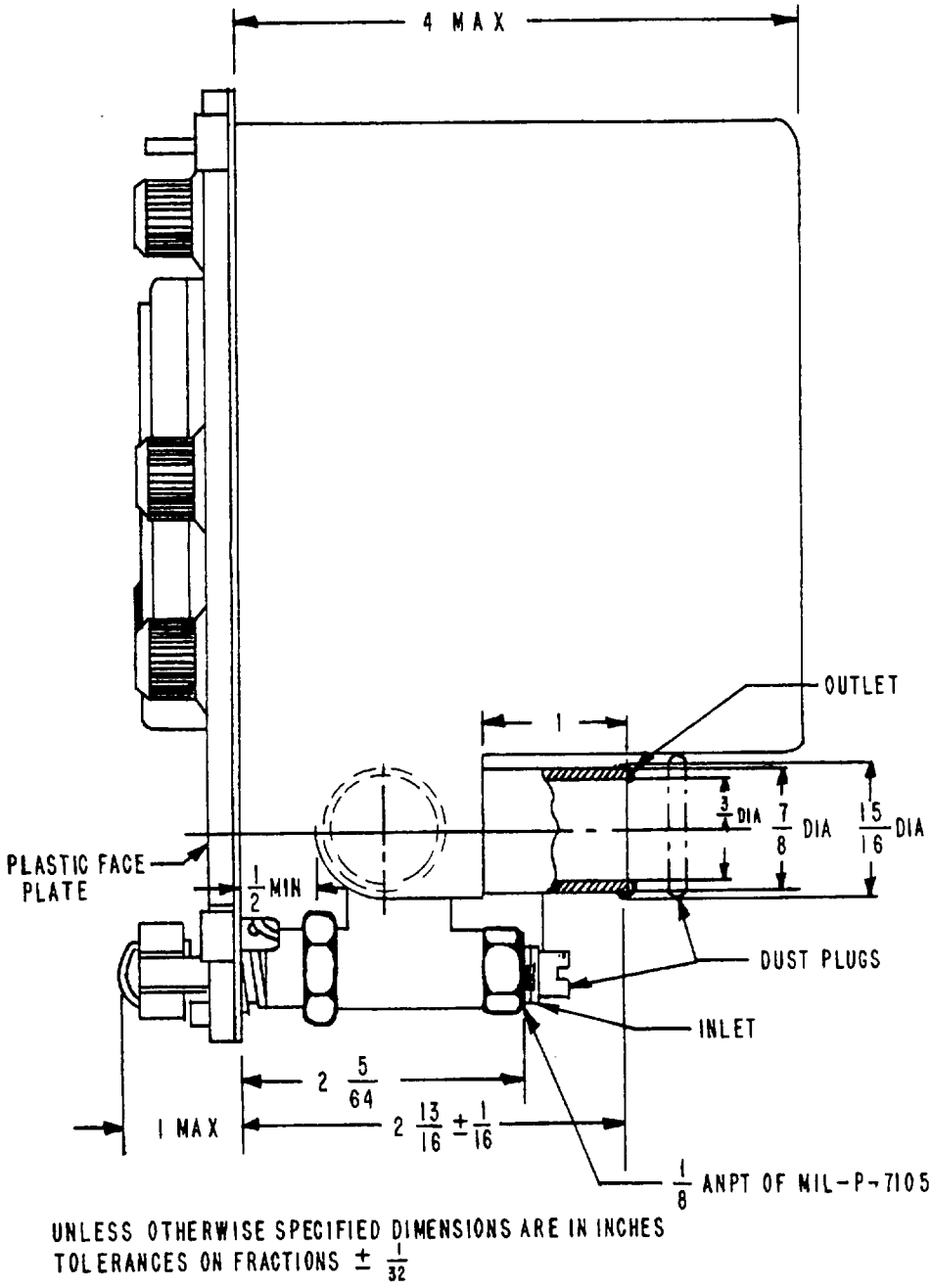


FIGURE 3. Regulator, Side View

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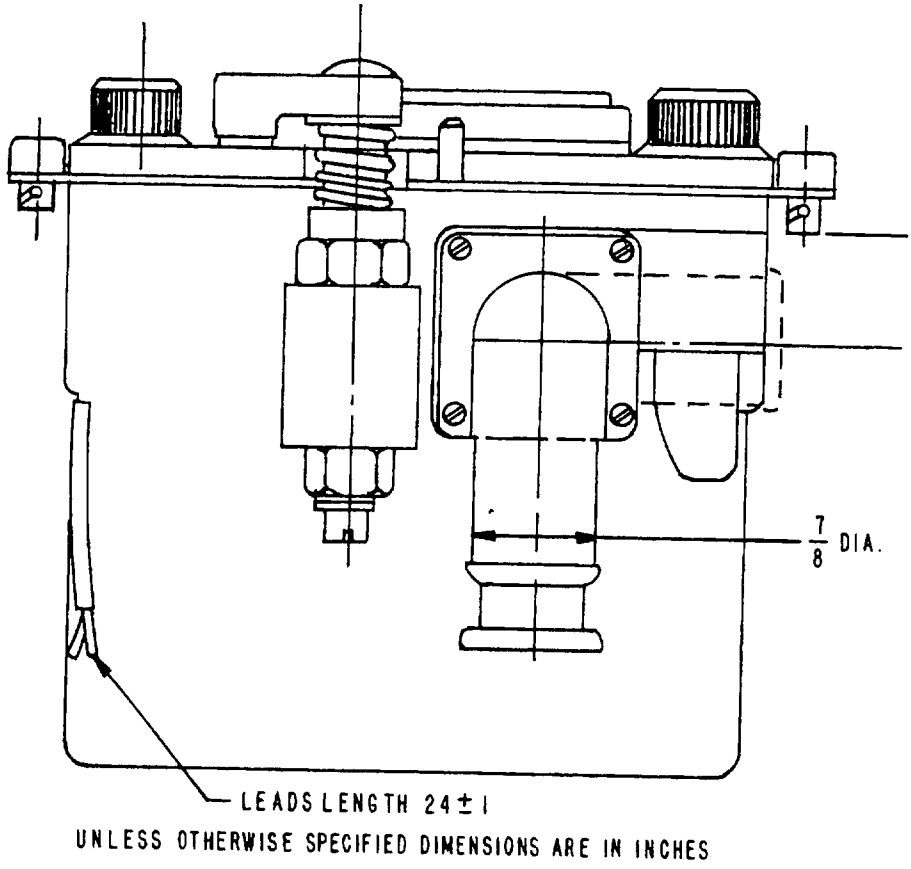


FIGURE 4. Regulator, Bottom View

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either to the right or to the left, it shall remain in position to supply outlet pressures. When the toggle is pushed straight in, a greater pressure shall be delivered at the outlet of the regulator; when released from the "straight in" position, the toggle shall return to its original position.

3.5.4 Lighted panel. The lighted panel shall conform to type II of MIL-P-7788. The lights shall conform to MS25453-1. The panel shall operate satisfactorily when tested as specified in 4.6.14. The wiring shall conform to figure 5.

3.5.5 Oxygen inlet filter. An oxygen filter at the oxygen inlet shall prevent foreign particles from entering the regulator. The oxygen inlet filter shall be located upstream from the oxygen supply valve. The oxygen inlet filter shall be equal to or interchangeable with grade 3 or grade 4 sintered bronze manufactured by the Moraine Products Company, Dayton, Ohio.

3.5.6 Oxygen supply valve. A valve at the inlet of the regulator shall shut off the oxygen supply when the regulator is not in use. The control knob of this valve shall be mounted so that it can be turned off and on at the regulator panel. The valve handle and marking shall be in accordance with figure 1. The valve shall be removable. Tension on the valve handle shall be great enough to prevent any change in setting because of vibration.

3.5.7 Flow indicator and pressure gage. A combination pressure gage and flow indicator shall be provided within the regulator and shall be mounted in the center of the regulator panel as shown in figure 1. The range of this pressure gage shall be 0 to 500 pounds per square inch. The dial shall conform to figure 6 and 3.5.7.1. Performance and general construction requirements of the gage shall be in accordance with MIL-G-25728. The flow indicator shall be designed so that the mechanism is actuated by flow from the regulator and shall not be required to operate at pressures greater than 17 inches of water. The flow indicator shall give a visual indication when flow is being drawn from the regulator.

3.5.7.1 Dial. The dial shall be 2-3/8 inches in diameter. All letters and numerals shall conform to MS33558. All graduations, numerals, and letters shall be white on a durable dull black.

3.5.8 Relief valve. A relief valve in the body of the regulator shall relieve any excessive pressure buildup at the outlet of the regulator. This valve shall vent through the regulator case to the atmosphere.

3.6 Performance

3.6.1 Excess flow characteristics. With an inlet pressure of 500 pounds per square inch, with no connection on the outlet of the regulator, and with the emergency toggle pushed straight in for a period of 3 seconds, there shall be no consistent mechanical noise or vibration within the regulator.

3.6.2 Flow suction characteristics. When either increasing or decreasing suctions are applied at the outlet to obtain the flows indicated in table I at altitudes from 0 to 25,000 feet and with the diluter at the"normal oxygen" and the "100 percent oxygen" position, the outlet suctions causing these

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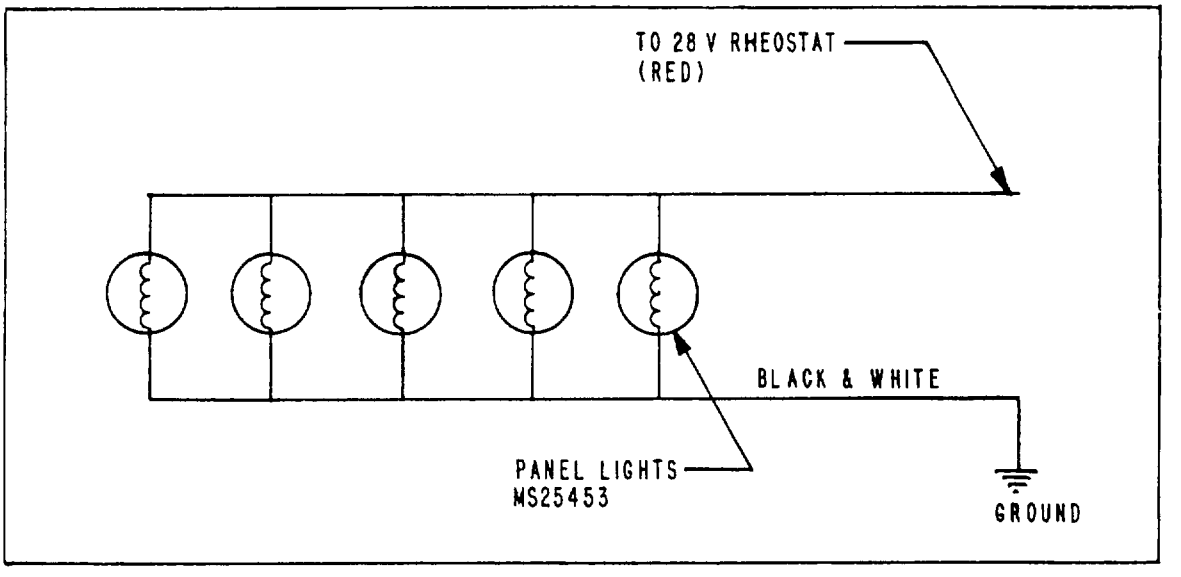


FIGURE 5. Wiring Diagram of Panel Lights

TABLE I. Flow Suction Characteristics

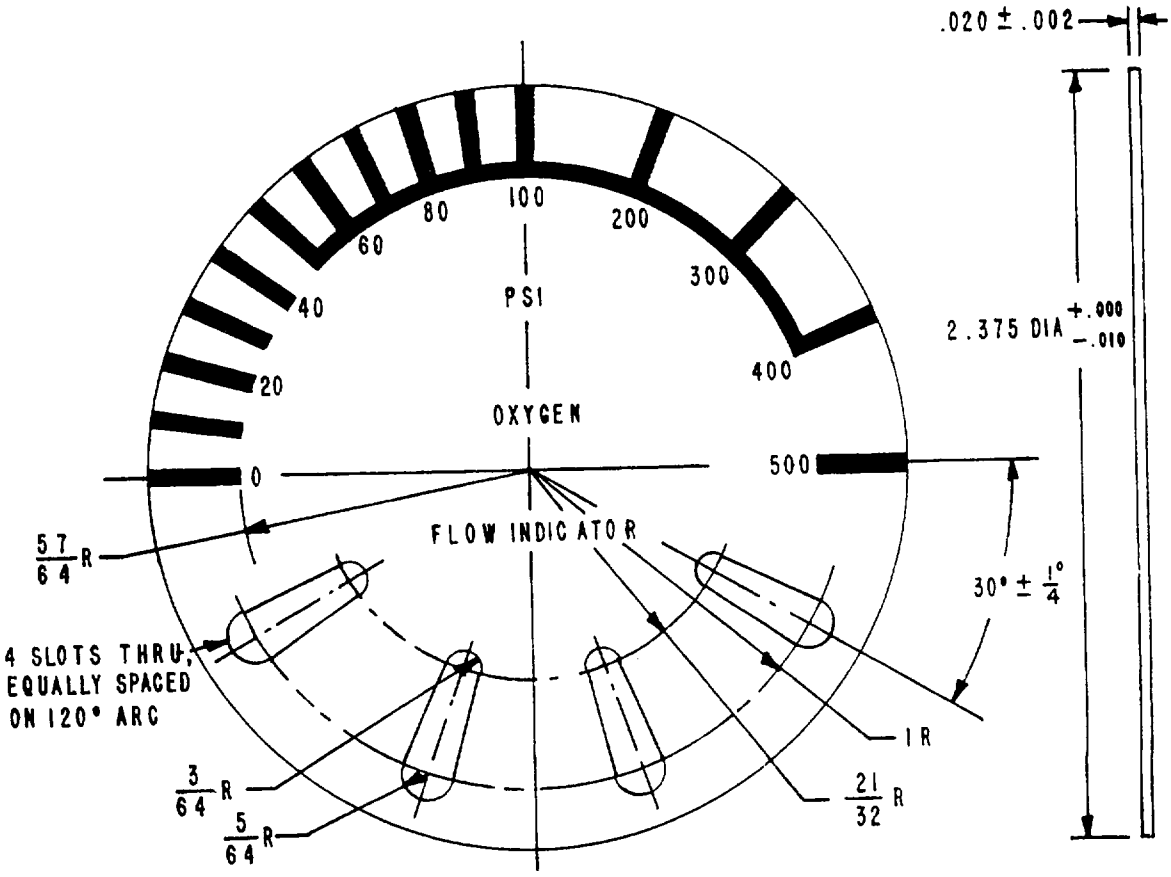
Supply Oxygen Pressure Pounds Per Square Inch	Flow of Oxygen or Oxygen-Air Mixture (Ambient LPM)	Maximum Outlet Suction (Inches of Water)
50 to 500	0 to 30	0.35
50 to 500	60	0.7
150 to 500	90	0.8

flows shall not exceed the values specified in table I. When suction is reduced to zero at altitudes from sea level to 20,000 feet, the flow shall not exceed 0.01 liter per minute (LPM).

3.6.3 Oxygen ratio. The ratio by volume of cylinder oxygen delivered to the total gas delivered by the regulator at various altitudes with any supply pressure in the range of 50 to 500 pounds per square inch shall be in accordance with table II.

3.6.4 Safety pressure and pressure breathing characteristics. With a supply pressure of 50 to 500 pounds per square inch gage, with the lever in the "normal oxygen" or "100 percent oxygen" position, and with either increasing or decreasing flows,

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GRADUATION: $\frac{17}{64}$ LONG BY .050 LINE.
NUMERALS 0 THRU 500: .125 ± .003 HIGH.
PSI, OXYGEN, FLOW INDICATOR: .093 ± .003 HIGH
GRADUATION 0 THRU 100 SHALL BE EQUALLY SPACED ON 90° ARC AS SHOWN AND GRADUATIONS 100 THRU 500 SHALL BE EQUALLY SPACED ON 90° ARC AS SHOWN.
TOLERANCE SHALL NOT EXCEED 0° 15' BETWEEN TWO LINES.
INSIDE SURFACES OF SLOTS SHALL BE DURABLE DULL BLACK.

UNLESS OTHERWISE SPECIFIED DIMENSIONS ARE IN INCHES
TOLERANCES:
FRACTIONS ± $\frac{1}{16}$
DECIMAL ± .005
ANGLES ± $\frac{1°}{2}$

FIGURE 6. Flow Indicator and Pressure Gage Dial

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TABLE II. Oxygen Ratio

Altitude (Thousand Feet)	Added Oxygen, Percent of Total Volume Flow (LPM)			
	1-14	15-50	51-85	86-135
0	0-100	0-30	0-30	0-50
5	1-100	1-33	1-33	1-50
10	6-100	6-45	6-35	6-60
15	14-100	14-52	14-50	14-70
20	24-100	24-55	24-65	24-80
25	40-100	40-80	40-80	40-90
28	60-100	60-100	60-100	60-100
32	98-100	98-100	98-100	98-100
	With Diluter Demand at "100 Percent Oxygen"			
All Altitudes	98-100	98-100	98-100	98-100

the pressure flow characteristics shall fall within the limits specified in table III and figure 7.

3.6.5 Relief valve. The relief valve shall vent a flow of at least 45 LPM at a pressure not greater than 3 inches of mercury. The relief valve shall vent through the regulator case to the atmosphere.

3.6.5.1 Vent. A suitable means shall be provided to vent any excessive first stage pressure.

3.6.6 Inward leakage. With the oxygen supply shut off and with a suction of 10 inches of water applied at the outlet as specified in 4.6.7, the leakage of air through the regulator at sea level shall be less than 0.2 LPM.

3.6.7 Over-all leakage. Over-all leakage of the regulator shall not exceed 0.01 LPM when 500 pounds per square inch of pressure is applied, as specified herein, to the inlet of the regulator.

3.6.8 Outward leakage. With any pressure in the range of 0 to 17 inches of water applied at the outlet of the regulator, the leakage through the

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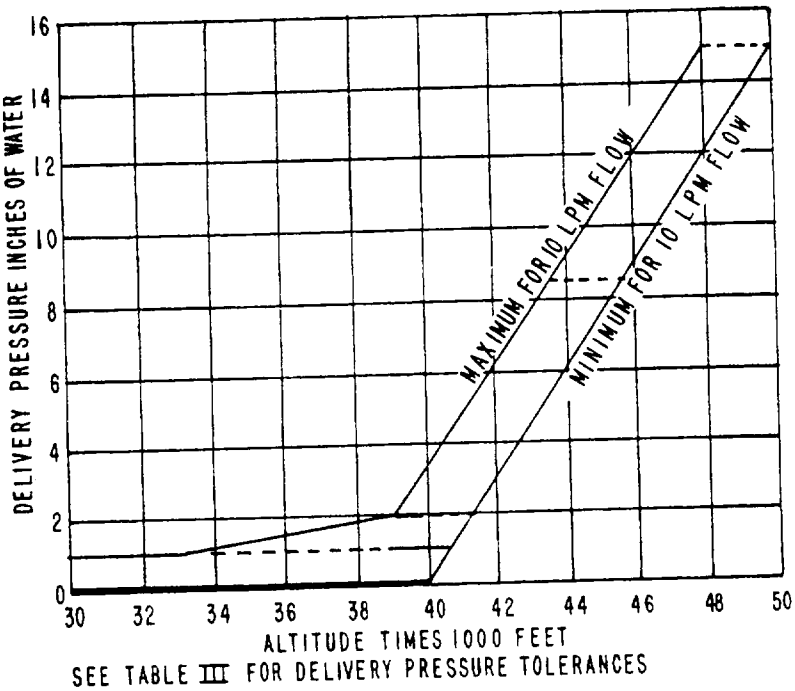


FIGURE 7. Safety Pressure and Pressure Breathing Limits

TABLE III. Delivery Pressure Tolerances

Range of Outlet Pressure 1/	Pressure Decreases (Inches of Water) from 10 LPM for a Flow of 135 LPM	Pressure Increases (Inches of Water) from 10 LPM for Flow of Zero LPM
0.0 to 1.0		.8
0.0 to 2.0	1.00	
2.0 to 15.0		1.25
8.5 to 15.0	1.25	

1/ Delivery pressure ranges in this column shall be allowed the tolerances specified in the second and third column for the stated flows.

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regulator (including the relief valve) shall not exceed 0.12 LPM.

3.6.9 Outlet leakage. With a pressure of 150 pounds per square inch applied at the inlet of the regulator and with the oxygen supply valve open, the leakage at the outlet of the regulator shall not exceed 0.01 LPM.

3.6.10 Emergency pressure. A pressure of 150 pounds per square inch shall be applied at the regulator inlet with the oxygen supply valve open. At a flow of 10 LPM and with the emergency toggle pushed either to the right or left, the pressure at the outlet of the regulator shall be increased 3.5 inches (plus or minus 0.50 inch) of water. At a flow of 90 LPM and with the diluter level in the "100 percent oxygen" position, the pressure shall not be less than 0.95 inch of water. When the emergency toggle is pushed straight in at a flow of 10 LPM, the regulator shall deliver an outlet pressure in the range of 10 to 17 inches of water.

3.6.11 Oxygen supply valve leakage. The oxygen supply valve shall show no sign of leakage when subjected to a pressure of 500 pounds per square inch gage as specified herein.

3.6.12 Flow indicator. With the diluter control lever in the "100 percent oxygen" position and the inlet pressure in the range of 50 to 500 pounds per square inch, a full indication of flow shall be given when a flow of 10 LPM is drawn from the regulator outlet at any altitude. When the diluter lever is in the "normal oxygen" position, a full indication of flow shall be obtained at a flow of 15 LPM at altitudes of 12,000 feet and above. At altitudes from sea level to 12,000 feet the same results shall be obtained when a flow of 25 LPM is drawn from the regulator. When the flow is reduced to zero at any altitude and with the diluter lever in either the "normal oxygen" or "100 percent oxygen" position, the flow indicator shall close immediately.

3.6.13 Panel light check. When panel lights are checked as specified in 4.6.14, all the light bulbs shall operate.

3.6.14 Overload. The regulator shall withstand a suction of 24 inches of water and a pressure of 27 inches of water applied for a 2-minute period at the outlet with the oxygen valve closed and with the diluter lever set at "100 percent oxygen" position. Venting of the second stage relief valve is permissible during the pressure phase of the test specified in 4.6.15.

3.6.15 Low temperature operation. While being subjected to a temperature of minus 65°F the regulator shall operate as specified herein.

3.6.16 High temperature operation. While being subjected to a temperature of plus 165 degrees Fahrenheit, the regulator shall operate as specified herein.

3.6.17 High temperature exposure. The regulator shall operate as specified herein after subjection to a temperature of plus 160 degrees Fahrenheit for a 7-hour period with 500 pounds per square inch of pressure at the inlet and with the oxygen supply valve open.

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3.6.18 Low temperature exposure. The regulator shall operate as specified herein after subjection to a temperature of minus 85 degrees Fahrenheit for a 48-hour period.

3.6.19 Reliability. The regulator shall have a minimum mean cycles between failures (MCBF) of 400,000 cycles with a minimum reliability of 0.97 for a 10-hour mission at a confidence factor of 0.90.

3.6.19.1 Longevity. The regulator shall have an operating life span (equipment longevity as defined in MIL-R-26667) of not less than 1,400,000 cycles before wearout failures occur or the equipment consistently fails to meet the specified MCBF index. Parts requiring replacement within this period of time and the normal operating period of such parts shall be reported by the contractor to the procuring activity.

3.6.20 Maintainability. The total mean active maintenance downtime, as defined in MIL-M-26512, in the maintenance of the regulator shall be the minimum possible as determined by maintainability verification as defined in 4.6.21.

3.7 Weight. The weight of the complete regulator shall not exceed 4 pounds and 14 ounces.

3.8 Threads

3.8.1 Screw threads. Unless otherwise specified in the thread for all machine screws 0.06 inch or larger in diameter shall conform to MIL-S-7742.

3.8.2 Pipe threads. Pipe threads shall be in accordance with MIL-P-7105.

3.9 Finish

3.9.1 The exterior of the regulator case shall be finished in dull black suitable for the purpose and comparable to high grade commercial finishes for this type of equipment.

3.9.2 Aluminum and aluminum alloy parts shall be anodized in accordance with MIL-C-5541 or MIL-A-8625.

3.10 Identification of product. Equipment, assemblies, and parts shall be marked in accordance with MIL-STD-130.

3.10.1 Date of manufacture. The date of manufacture, consisting of month and year, shall be legibly stamped with inspector's ink on the regulator in a place where it will not be defaced during usage. A suitable rubber stamp made in the form of a 1/2-inch square with a horizontal line dividing the square into two segments shall be used for this marking. The upper segment shall contain the number of the month, and the lower segment shall contain the last two numbers of the year.

3.10.2 Serial numbers. Serial numbers shall be assigned by the manufacturer. The serial number shall include a prefix consisting of AF, two numerals designating the fiscal year in which the regulators were delivered, and a dash (example: AF57-).

3.11 Part numbering of interchangeable parts. All parts having the same manufacturer's part number shall be functionally and dimensionally interchangeable. The item identification and part number requirements of MIL-D-70327 shall govern the manufacturer's part numbers and changes thereto.

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3.12 Workmanship. The regulators shall be manufactured in a thoroughly workmanlike manner. The finished regulators and all integral parts shall be clean and free of oils, greases, and any other materials that might adversely affect the operation of the regulator.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Classification of tests. The inspection and testing of regulators shall be classified as follows:

a. Preproduction tests

b. Acceptance tests.

4.3 Test conditions

4.3.1 Gas. The high pressure gas used in testing the regulator shall be oxygen conforming to type I of MIL-O-27210; water-pumped nitrogen conforming to type I, class 1, grade B of BB-N-411; or water-pumped air equivalent in dryness to BB-N-411 nitrogen. If nitrogen or air is used, appropriate density corrective factors shall be applied to the flowmeter used to correct the effect of the lower density gas on the flowmeter used and on the performance of the regulator.

4.3.2 Pressure and temperature. Unless otherwise specified, all tests shall be made at an atmospheric pressure of 28 to 32 inches of mercury, at a temperature of 77 degrees (plus or minus 18 degrees) Fahrenheit, and at a relative humidity of 80 percent or less. If the atmospheric pressure or the temperature is substantially different from these values, proper allowance shall be made for the difference from the specified conditions.

4.3.3 Adjusted pressure. Unless otherwise specified, the inlet oxygen pressure shall be adjusted to 500 pounds per square inch during all inspection tests.

4.3.4 Orientation. Unless otherwise specified, all tests of the regulator shall be conducted with the panel of the regulator in a horizontal (face up) plane.

4.4 Preproduction testing

4.4.1 Preproduction test samples tested by the contractor. The contractor shall subject two regulators to the preproduction tests specified in 4.4.3.

4.4.2 Preproduction test sample for the procuring activity. The contractor shall submit one regulator that has passed the preproduction tests to the procuring activity.

4.4.3 Preproduction tests. Preproduction tests shall consist of all the tests described under 4.6.

4.4.3.1 Preproduction reliability tests. The two preproduction samples specified in 4.4.1 shall be tested by the contractor for reliability and one of the two regulators shall be continued through a longevity test in accordance with MIL-R-26667. Duration of the MCBF

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demonstration test on each unit shall not be less than three times the specified MCBF. Acceptance, permitting production, shall be judged on the basis of the table entitled "Reliability Accept-Reject Criteria (Method 2)" of MIL-R-26667. Recording, data handling, and reporting procedures shall be in accordance with MIL-R-26667.

4.5 Acceptance tests. Acceptance tests shall consist of:

a. Individual tests

b. Sampling tests

4.5.1 Individual tests. Each regulator shall be subjected to the following tests as described under 4.6:

a. Examination of product

b. Excess flow characteristics

c. Flow suction characteristics

d. Oxygen ratio

e. Safety pressure and pressure breathing characteristics

f. Relief valve

g. Inward leakage

h. Over-all leakage

i. Outward leakage

j. Outlet leakage

k. Emergency pressure

l. Oxygen supply valve leakage

m. Flow indicator

n. Panel lighting check.

4.5.2 Sampling tests. During the sampling tests specified in 4.5.2.1 and 4.5.2.2, the regulator shall be mounted with the regulator panel in a vertical plane. With the panel vertical, outlet pressures of 1.5 inches of water greater than the maximum pressures specified in figure 7 shall be allowed at altitudes from 30,000 to 50,000 feet. During the sampling tests specified in 4.5.2.3, the regulator shall be mounted in the horizontal face downward position. With the panel facing downward, pressures of 3.0 inches of water greater than the maximum shall be allowed at altitudes from 30,000 to 50,000 feet.

4.5.2.1 Sampling Plan A. One regulator, selected at random from the first 25 produced from each lot of 100 regulators or fraction thereof, shall be subjected to the following tests as described under 4.6:

a. Overload

b. Low temperature operation

c. High temperature operation.

4.5.2.2 Sampling Plan B. Three regulators, selected at random from each lot of 500 regulators or fraction thereof, shall be subjected to the following tests as specified under 4.6:

a. High temperature exposure

b. Low temperature exposure

c. Reliability test level.

4.5.2.3 Sampling plan C. One regulator, selected at random from the

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regulators on each contract or order, shall be subjected to all of the tests specified under 4.6.

4.5.2.4 Rejection and retest for sampling plans A and B. When one or more items from a lot fail to meet the specification, acceptance of all items in the lot shall be withheld until the extent and cause of failure have been determined. The contractor shall explain fully to the Government representative the cause of failure and the action taken to preclude recurrence. After correction, all of the sampling tests shall be repeated.

4.5.2.5 Rejection and retest for sampling plan C. When an item selected from a production run fails to meet the specification no item still on hand or later produced shall be accepted until the extent and cause of failure have been determined and appropriately corrected. The contractor shall explain fully to the Government representative the cause of failure and the action taken to preclude recurrence. After correction, all of the tests shall be repeated.

4.5.2.6 Individual tests may continue. For production reasons, individual tests or sampling plans may be continued pending the investigation of a sampling test failure. But final acceptance of the entire lot or lots produced later shall not be made until it is determined that all items meet all the requirements of the specification.

4.5.4 Defects in items already accepted. The investigation of a test failure could indicate that defects may exist in items already accepted. If so, the contractor shall fully advise the procuring activity of all the defects likely to be found and the method of correcting them.

4.6 Test methods. Vibration, flutter, or chatter ocurring during any of the following tests shall be cause for rejection of the regulator. If the regulator vibrates during test and the vibration can be stopped by changing or varying the flow and returning to the test point, the regulator is acceptable. If vibration persists after several attempts or if vibration can be obtained by simulating a breathing cycle, the regulator shall not be acceptable.

4.6.1 Examination of product. Each regulator shall be examined to determine conformance to this specification with respect to material, workmanship, weight, and markings.

4.6.2 Excess flow characteristics. With an inlet pressure of 500 pounds per square inch and without any connections on the outlet, the emergency toggle shall be pushed straight in for 2 seconds. If there is a mechanical noise or vibration during this test, the test shall be repeated. If three succeeding tests fail to produce a mechanical noise or vibration, the regulator is acceptable.

4.6.3 Flow suction characteristics. The regulator shall be connected to a pressure source capable of producing the supply pressures indicated in table I, and the oxygen supply valve shall be opened. The inlet pressures shall be set in accordance with table I and flows indicated in table I shall be drawn from the regulator. With the diluter level in both the "normal oxygen" and "100 percent oxygen" positions, at altitudes of 25,000 feet and below, and with either increasing or decreasing flows, the suctions required to

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produce the flows specified in table I shall not exceed the suction values specified in table I. After the suction is reduced to zero at altitudes from sea level to 20,000 feet, the flow shall not exceed 0.01 LPM. All measurements of the outlet suction pressures shall be made directly at the outlet of the regulator.

4.6.4 Oxygen ratio. The inlet pressure shall be set to 150 pounds per square inch and the ratio, by volume, of cylinder oxygen delivered to the total gas delivered by the regulator at various altitudes shall be measured at room temperature and under conditions acceptable to the procuring activity to determine compliance with the requirements of table IV.

4.6.5 Safety pressure and pressure breathing characteristics. The inlet oxygen pressure shall be set at 150 pounds per square inch gage with the oxygen supply valve open and the diluter lever in the "normal oxygen" position. At the delivery pressure specified in the first column of table V corresponding to a particular flow, the regulator shall be within the altitude range specified in the second column of table V.

4.6.6 Relief valve. The diluter lever shall be set at the "100 percent oxygen" position, and increasing pressure shall be applied in the outlet of the regulator through a rotometer or any other flow measuring device. The pressure required to vent 45 LPM through the relief valve shall not be greater than 3 inches of mercury.

4.6.7 Inward leakage. With the diluter lever set at the "100 percent oxygen" position and with the oxygen supply valve closed, a suction of 10 inches of water shall be applied at the outlet of the regulator. Leakage shall be measured by a flowmeter connected at the outlet and shall be less than 0.2 LPM.

4.6.8 Over-all leakage. A pressure of 500 pounds per square inch gage shall be applied at the inlet of the regulator with the oxygen supply valve open. The oxygen supply valve shall then be closed, locking the 500 pounds per square inch gage pressure in the regulator. The oxygen supply line shall be disconnected with the oxygen supply valve remaining in the closed position. The regulator pressure gage shall then be observed for a drop in pressure indicating leakage. The allowable leakage shall not exceed 0.01 LPM, determined by the pressure drop over a period of 2 minutes. The allowable pressure drop in pounds per square inch gage shall be calculated from the known volume of the system, including the pressure gage.

4.6.9 Outward leaking. With the oxygen supply valve closed, a pressure of 17 inches of water shall be applied at the outlet of the regulator through a flow-measuring device such as a rotometer. This 17 inches of water pressure shall be maintained, and the flow through the measuring device necessary to maintain this pressure shall not exceed 0.12 LPM. During this test the relief valve shall not be covered since the allowable leakage through this valve at 17 inches of water is included in the given leakage value of 0.12 LPM.

4.6.10 Outlet leakage. A pressure of 150 pounds per square inch gage shall be applied at the regulator inlet with the oxygen supply valve open. A test conducted at room temperature shall be made, using

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TABLE IV. Oxygen Ratio Tests

Altitude (Thousand Feet)	Outlet Flow (LPM)	Percent Oxygen Added	
		Minimum	Maximum
10	15	6	45
10	50	6	35
10	85	6	35
15	5	14	100
15	15	14	52
15	50	14	42
20	15	24	55
20	50	24	55
20	85	24	65
25	5	40	100
25	15	40	80
25	50	40	80
32	85	98	100

leak test compound conforming to MIL-L-25567, by drawing a film across the outlet fitting of the regulator. The test shall be repeated at a temperature of minus 65 degrees Fahrenheit, using the regulator test stand used in conduction of the oxygen ratio tests. The outlet flow valve of the stand should be closed, and an increase in outlet pressure shall indicate leakage. Outlet leakage shall not exceed 0.01 LPM.

4.6.11 Emergency pressure. A pressure of 150 pounds per square inch gage shall be applied at the regulator inlet with the supply valve open. A flow of 10 LPM shall then be drawn from the regulator. With the emergency toggle pushed either to the right or left, the pressure recorded at the outlet of the regulator shall be increased by 3.5 inches (plus or minus 0.5 inch) of water. At a flow of 90 LPM and with the diluter lever in the "100 percent oxygen" position, the pressure shall not be less than 0.95 inch of water. With an outlet flow of 10 LPM, the regulator shall deliver an outlet pressure in the

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TABLE V. Safety Pressure And Pressure-Breathing Characteristics

Outlet Pressure for 10 LPM Flow (Inches of Water)	Altitude Range (Thousand Feet) for Column I	Maximum Outlet Pressure Increase for Zero LPM (Inches of Water)	Maximum Outlet Pressure Decrease for Indicated Flow (Inches of Water)
1.0	27 to 40	1.6	0.9 for 70 LPM
2.0	29 to 41	1.3	1.3 for 135 LPM
8.5	42 to 45	1.3	1.3 for 135 LPM
15.0	47 to 50	1.3	1.3 for 135 LPM

NOTE: The outlet pressure for the altitude range of 30,000 feet to 38,000 feet shall not fall below 0.01 inch of water for flow of 0 to 10 LPM. The outlet pressure for the altitude range of 34,000 feet to 40,000 feet shall not fall below 0.01 inch of water for flow of 0 to 70 LPM.

range of 10 to 17 inches of water when the emergency toggle is pushed straight in.

4.6.12 Oxygen supply valve leakage. A pressure of 500 pounds per square inch gage shall be applied at the regulator inlet with the oxygen supply valve closed and the emergency toggle shall be pushed either to the right or to the left. The regulator shall then be tested for leakage at the outlet, as described in 4.6.10, using the test stand. Any leakage shall be cause for rejection.

4.6.13 Flow indicator. With an inlet pressure of 150 pounds per square inch gage, the operation of the flow indicator shall be checked. At sea level the flow indicator shall be fully opened at 25 LPM with the diluter lever in the "normal oxygen" position and at 10 LPM with the diluter lever in the "100 percent oxygen" position. At an altitude sufficient to give 17 inches of water delivery pressure, the flow indicator shall be fully opened at an ambient flow of 10 LPM. When the flow is reduced to zero, the flow indicator shall close immediately.

4.6.14 Panel lighting check. The panel shall be connected to a 28-volt power source and visually examined to see if all the light bulbs operate.

4.6.15 Overload. With the oxygen supply valve closed and with the diluter in the "100 percent oxygen" position, a pressure of 27 inches of water and then a suction of 24 inches of water shall be applied at the outlet of the regulator for a period of 2 minutes each. These tests shall not be detrimental to the operation of the regulator or to its components. Any failure of the regulator to withstand the suction and pressure shall be cause for rejection.

4.6.16 Low temperature operation. The regulator shall be subjected to

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an inlet pressure of 150 pounds per square inch gage and shall be stabilized at a temperature of minus 65 degrees Fahrenheit. While maintained at the specified temperature, the regulator shall be subjected to cyclic operation, simulating service use, for a period of 2 minutes. The cyclic operation shall consist of approximately 10 cycles per minute at a flow from 0 to 70 LPM and back to zero LPM. Following the cyclic operation and while still at minus 65 degrees Fahrenheit, the regulator shall be subjected to the tests specified in 4.6.3, 4.6.4, and 4.6.5.

4.6.17 High temperature operation. Except that the temperature shall be plus 165 degrees Fahrenheit instead of minus 65 degrees Fahrenheit, the regulator shall be subjected to the test specified in 4.6.16.

4.6.18 High temperature exposure. The regulator shall be subjected to a temperature of plus 160 degrees Fahrenheit for a period of 7 hours, while maintaining an inlet pressure of 500 pounds per square inch gage and with the oxygen supply valve open. The regulator shall then be allowed to reach room temperature and shall be subjected to the test specified in 4.6.16.

4.6.19 Low temperature exposure. The regulator shall be subjected to a temperature of minus 85 degrees for a period of 48 hours. After the 48-hour period, the regulator shall be allowed to return to room temperature and shall be subjected to the tests specified in 4.5.1.

4.6.20 Reliability test level. The reliability test level shall consist of simulated breathing cycles with oxygen as the test fluid, while the regulator is subjected to vibration conditions. For 50 percent of the breathing cycles, the diluter lever shall be in each of the "100 percent oxygen" and the "normal oxygen" positions. The peak delivery rate for 90 percent of the breathing cycles (both positions of diluter lever) shall be 30 liters per minute, and for the remaining 10 percent of the breathing cycles shall be 70 liters per minute. The vibration to which the regulator is subjected during this test shall be at a double amplitude of 0.018 to 0.020 inch. The frequency of vibration shall vary regularly from 300 to 3,000 cycles per minute and back to 300 cycles per minute: the rate of change in frequency shall be constant. The regulator shall be so mounted that the panel of the regulator is horizontal during this test. A failure shall be identified as any malfunction that causes improper delivery of concentration or pressure of oxygen from the regulator or improper indication of oxygen flow from the regulator. To determine if a failure has ocurred the regulator shall be checked during reliability testing as follows: Flow indicator observations and regulator outlet pressure measurements shall be made at least five times every 24 hours. The regulators shall be tested for oxygen ratio and pressure breathing characteristics at least once each 400,000 cycles of regulator operation. Except for lamps in the lighting portion of the regulator, no parts shall be replaced as preventive maintenance during reliability testing.

4.6.21 Maintainability verification. Verification of the maintainability requirements for the regulator shall be in accordance with the paragraph entitled "Maintenance Task Test and Demonstration" of Appendix A of MIL-M-26512. As a minimum, the following

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field equipment, or equivalent shall be considered in the time motion studies of maintenance task simulation: Type TTU-213/E tester covered by MIL-T-38180.

5. PREPARATION FOR DELIVERY

5.1 Preservation and packaging

5.1.1 Level A. Regulators shall be preserved in accordance with method IA8 of MIL-P-116 and shall be packed in boxes conforming to PPP-B-566, PPP-B-636, PPP-B-665 or PPP-B-676.

5.1.2 Level C. Regulators shall be preserved and packaged in accordance with the manufacturer's commercial practice.

5.2 Intermediate packaging. Only identical items shall be included in an intermediate package. Requirements for intermediate packaging for items preserved and packaged in accordance with 5.1.2 shall be at the option of the contractor. Regulators preserved and packaged in accordance with 5.1.1 shall be packaged in boxes conforming to PPP-B-566, PPP-B-636, PPP-B-665, and PPP-B-676. Unless otherwise specified by the procuring activity, the quantity of unit packages to be included in each intermediate package shall be at the option of the contractor and as governed by the limitations of the container being used.

5.3 Packing

5.3.1 Level A. Regulators preserved and packaged in accordance with 5.1.1 and 5.2 shall be packed in exterior type shipping containers conforming to PPP-B-591; PPP-B-601; PPP-B-621; or type CF or SF, class domestic of PPP-B-636. As far as practical, exterior containers shall be of uniform shape and size, shall be of minimum cube and tare consistent with the protection required, and shall contain identical quantities. The gross weight of each pack shall not exceed the maximum weight limitations of the container selected. Containers shall be closed and strapped in accordance with the applicable container specification or appendix thereto. Containers shall be provided with a case liner conforming to MIL-L-10547 and shall be sealed in accordance with the appendix thereto. The case liner will not be required when the unit, the intermediate, or the exterior container conforms to type CF or SF, class domestic and is sealed at all joints and seams, including the manufacturer's joint, with tape conforming to PPP-T-60.

5.3.2 Level B. Regulators preserved and packaged in accordance with 5.1.1 and 5.2 shall be packed in domestic type exterior containers conforming to PPP-B-585, PPP-B-591, PPP-B-601, PPP-B-621, or PPP-B-636. Exterior containers shall be of minimum cube and tare consistent with the protection required. As far as practical, exterior containers shall be of uniform shape and size and shall contain identical quantities. The gross weight of each pack shall not exceed the maximum weight limitations of the container selected. Containers shall be closed and strapped in accordance with the applicable container specification or appendix thereto. When fiberboard containers are used, the fiberboard shall meet the special requirements table of PPP-B-636, as applicable.

5.3.3 Level C. Packages which require overpacking for acceptance by the carrier shall be packed in exterior type shipping containers in a manner that will insure safe

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transportation at the lowest rate to the point of delivery. Containers shall meet Consolidated Freight Classification Rules or regulations of other common carriers as applicable to the mode of transportation.

5.4 Physical protection. Cushioning, blocking, bracing, and bolting as required shall be in accordance with MIL-STD-1186 except that for domestic shipment, waterproofing requirements for cushioning materials and containers shall be waived. Drop tests of MIL-STD-1186 shall be waived when preservation, packaging, and packing of the item are for immediate use or when drop tests of MIL-P-116 are applicable.

5.5 Marking. Interior and exterior containers shall be marked in accordance with MIL-STD-129 and as follows:

If in storage after ____*____ the regulator shall be turned in as a repairable item.

*Storage shall be 24 months from date of inspection if a metal container is used and 18 months if a container other than metal is used.

5.5.1 Precautionary marking. Two opposite sides of each interior container shall be marked with the following precautionary marking:

FRAGILE - HANDLE WITH CARE.

5.5.2 Date marking. The date all containers were manufactured shall be stamped on the regulator. The stamp specified in 3.10.1 shall be used for this marking.

6. NOTES

6.1 Intended use. The regulator covered by this specification is intended to be used in aircraft at altitudes up to 50,000 feet.

6.2 Ordering data. Procurement documents should specify the

a. Title, number, and date of this specification

b. Preproduction requirements

c. Selection of applicable levels of preservation and packaging and packing

d. Unit quantities.

Custodian:
Air Force - 11

Preparing activity:
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Other interests:
Reviewers:
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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 119-R004

INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity *(as indicated on reverse hereof)*.

SPECIFICATION

ORGANIZATION *(Of submitter)* | CITY AND STATE

CONTRACT NO. | QUANTITY OF ITEMS PROCURED | DOLLAR AMOUNT
$

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES.

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO IF "YES", IN WHAT WAY?

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity)* | DATE

DD, FORM 1 APR 63 1426 REPLACES NAVSHIPS FORM 4863, WHICH IS OBSOLETE C-6278

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